

CMSC 218 Review

Dylan Tang

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1 Pandas

- **Group by:** Collapses rows that share the same attribute into one row, and performs an aggregate operation on the collapsed rows.
- Ex. `df.groupby(["col"]).mean()`
- To group with multiple columns, use `agg()`:
`df.agg(col1_mean = ("col1", "mean"), col2_median = ("col2", "median"))`
- **Merge:** Merge by a common column
- Ex. `merged = pd.merge(df1, df2, on="id", how="inner")`
- how:
 - `inner` - merge takes intersection of both `df1["id"]`, `df2["id"]`
 - `outer` - merge takes union of `df1["id"]`, `df2["id"]`
 - `left` - merge uses `df1["id"]` as index
 - `right` - merge uses `df2["id"]` as index
- `outer`, `left`, `right` can produce NaN.

Example (Group by Departments)

```
df.groupby(["Department"])["Salary"].mean()
```

- The above line can be read as: group by department, and take the mean of each department's salary.
- More generally, `groupby` takes on the following form:

```
df.groupby([group_var])[value_var].agg_func()
```

Example (Pandas Merge with Duplicate Keys)

```
df1 = pd.DataFrame({"id": [1, 1, 2, 3], "dept": ["sales", "VP", "tech", "hr"]})
df2 = pd.DataFrame({"id": [1, 2, 2, 3], "hrs": [10, 20, 30, 15]})
df1.merge(df2)
```

The above merged df has the following form:

id	dept	hrs
1	sales	10
1	VP	10
2	tech	20
2	tech	30
3	hr	15

2 Sampling

- Given an RV X , $\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$
- **68-95-99.7 rule** - 68% of samples fall within 1 SD, 95% of samples fall within 2 SD, 99.7% of samples fall within 3 SD
- **Stratified samples** - point estimate is a weighted average of sub populations
 - $\bar{X} = w_1\bar{X}_1 + \dots + w_n\bar{X}_n$
 - $\text{Var}(\bar{X}) = \text{Var}(w_1\bar{X}_1) + \dots + \text{Var}(w_n\bar{X}_n) = w_1^2\text{Var}(\bar{X}_1) + \dots + w_n^2\text{Var}(\bar{X}_n)$
- Ensures representativeness of all populations according to preset weights
- **95% CI for proportions:**

$$C = \hat{p} \pm 1.96 \cdot \text{SE}(\hat{p})$$

$$\leq \hat{p} \pm 1.96 \sqrt{\frac{0.5(0.5)}{n}}$$

3 Linear Model

- $Y = mX + b + \varepsilon$
- $\mathbb{E}[Y] = m\mathbb{E}[X] + b$
- $\text{Var}(Y) = m^2\text{Var}(X) + \text{Var}(\varepsilon)$
- **Signal-Noise Ratio:**

$$\text{SNR} = \frac{m^2\text{Var}(X)}{\text{Var}(\varepsilon)}$$

- $\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$
- Affected by scaling: $\text{Cov}(aX, Y) = a \cdot \text{Cov}(X, Y)$
- Measures the **scale** of association between X and Y
- **Correlation:**

$$\rho = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

- Not affected by scaling $X \sim Y$
- Measures the **strength**
- **1-D OLS:**

$$\hat{m} = \frac{\text{Cov}(X, Y)}{\text{Var}(X)} \Rightarrow \text{if } X, Y \text{ standardized, } \hat{m} = \frac{\text{Cov}(X, Y)}{\text{Var}(X)} = \text{Cov}(X, Y)$$

$$\hat{b} = \mathbb{E}[Y] - \hat{m}\mathbb{E}[X]$$

$$\rho = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} = \text{Cov}(X, Y) \quad X, Y \text{ standardized}$$

$$\Rightarrow \hat{m} = \rho$$

4 Bias-Variance Decomposition

$$\mathbb{E}[(\hat{y} - y)^2] = \underbrace{(\mathbb{E}[\hat{m}] - m)^2}_{\text{Bias}^2(\hat{m})} + \underbrace{(\mathbb{E}[\hat{b}] - b)^2}_{\text{Bias}^2(\hat{b})} + \underbrace{\text{Var}(\hat{m}) + \text{Var}(\hat{b})}_{\text{Variance}} + \underbrace{\text{Var}(\varepsilon)}_{\text{Error}}$$

5 Multivariate OLS

- $y = X\beta + \varepsilon$, $X \in \mathbb{R}^{n \times p}$, $\beta \in \mathbb{R}^{p \times 1}$, $\varepsilon \in \mathbb{R}^{n \times p}$, $y \in \mathbb{R}^{n \times p}$
- $\arg \min_{\beta} |Y - X\beta|^2 \Rightarrow \hat{\beta} = (X^T X)^{-1} X^T y \Rightarrow \hat{y} = X\hat{\beta} = X(X^T X)^{-1} X^T y$
- $\beta_j \neq \frac{\text{Cov}(X_j, y_j)}{\text{Var}(X_j)}$, so $\hat{m} \neq \rho$ after standardizing X, y .
- A large β_j only indicates that one predictor is important in context of other predictors. To compare β_i, β_j , X_i, X_j must be standardized.
- $R^2 = 1 - \frac{\sum (\hat{y} - y)^2}{\sum (\bar{y} - y)^2} = 1 - \frac{\text{SSE}}{\text{SST}}$
- Measures how much better we do than the baseline model, $\bar{y}$.

5.1 Ridge Regression

Smooth predictions by adding L2 regularizer:

$$\arg \min_{\beta} |Y - X\beta|^2 + \alpha |\beta|^2 \Rightarrow \hat{\beta} = (X^T X + \alpha I)^{-1} X^T y \Rightarrow \hat{y} = X(X^T X + \alpha I)^{-1} y$$

- Trades off efficiency (how much the model relies on the data) for robustness (how generalizable the data is)
 - Trades off a small increase in bias² = $(\mathbb{E}[\hat{w}] - w)^2$ for a large decrease in $\text{Var}(\hat{w})$

6 Building a Regression Predictor

- **Inference model:** Given (x_i, y) pairs, produce $f(\theta)$.
- **Prediction model:** Given X, y as inputs, organizes the inputs in (X, y) pairs.

Typical regression pipeline:

```
X = df[...]
y = df[...]
X = StandardScaler().fit_transform(X)
X_train, X_test, y_train, y_test = train_test_split(X, y)

model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
r2_score(y_test, y_pred)
```

7 Generalization

- T_e - expected testing loss
- T_r - expected training loss
- $T_e - T_r = G$ (generalization gap)

$$\underbrace{(T_e - T_r)}_G + T_r = T_e$$

- A complex model is well-fit to the training data, and as such it is less generalizable - $G \uparrow$, $T_r \downarrow$.
- A simple model likely can be well generalized, but has high training error - $G \downarrow$, $T_r \uparrow$.
- A good model jointly minimizes G , T_r .

This tradeoff is also seen in the Bias-Variance Decomposition for squared loss:

$$\mathbb{E}[\ell(y_{\text{out}}, y_{\text{true}})] = \mathbb{E}[\ell(\hat{f}(x), y_{\text{true}})]$$

$$\mathbb{E}[(y_{\text{true}} - \hat{f}(x))^2] = \underbrace{(\mathbb{E}[\hat{f}(x)] - y_{\text{true}})^2}_{\text{Bias}} + \underbrace{\text{Var}(\hat{f}(x))}_{\text{Variance}} + \underbrace{\sigma^2}_{\text{Noise}}$$

- A simple model has less variance, but more bias
- A complex model has low bias, high variance.
- OLS jointly optimizes over bias and variance by minimizing squared loss.

8 Autoregression

- **Autoregression:** Using past values of y to predict future values of y .
- Any model that predicts an outcome based on time can may be subject to autoregression.
- **AR(1) Model:**

$$y_t = \underbrace{\beta_{\text{lag}} y_{t-1}}_{\text{AR(1) param}} + \underbrace{X\beta}_{\text{other covariates}}$$

Typical AR(1) Pipeline:

```
df["y_{t-1}"] = df["y"].shift(1)
df.dropna(how="any")

X = df[...]
y = df[...]

tscv = TimeSeriesSplit()
for train_idx, test_idx in tscv.split(X):
    X_train, X_test = X[train_idx], X[test_idx]
    y_train, y_test = y[train_idx], y[test_idx]

model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
r2_score(y_test, y_pred)
```

- **Longitudinal model** - uses $y_{t-k}, \dots, y_{t-1}$ to predict y_t

- **Cross-sectional model** - uses covariates X to predict y_t

Example (Sales)

A linear regression predictor has been fit on data that records sales in the given month and year.

What are the potential issues in the model fitting pipeline below?

```
X = df[["month", "year"]]
y = df[["sales"]]
X_train, X_test, y_train, y_test = train_test_split(X, y)

# Train the model
linear_model = LinearRegression()
linear_model.fit(X_train, y_train)

# Evaluate the model
y_pred = linear_model.predict(X_test)
r2_score(y_test, y_pred)
```

Issues:

- Model does not account for autoregressive nature of data. Given that the predictors are w.r.t time, the best predictor for this month's sales may actually be last month's sales.
- The predictors are not standardized, meaning that we cannot compare the magnitude of coefficients directly. Thus, we cannot isolate which predictor in the model is most significant.

A revised implementation would be:

```
# Create the AR(1) column
df["prev_month_sales"] = df["sales"].shift(1)
df.dropna(inplace=True)

X = df[["month", "year", "prev_month_sales"]]
y = df[["sales"]]

# Standardize X, y
X = StandardScaler().fit_transform(X)
y = StandardScaler().fit_transform(y)

# Use tscv.split since one of our predictors are time-dependent
tscv = TimeSeriesSplit()
for train_idx, test_idx in tscv.split(X):
    X_train, X_test = X[train_idx], X[test_idx]
    y_train, y_test = y[train_idx], y[test_idx]

# Train the model
linear_model = LinearRegression()
linear_model.fit(X_train, y_train)

# Evaluate the model
y_pred = linear_model.predict(X_test)
r2_score(y_test, y_pred)
```

9 Logistic Regression

Given (X_i, y_i) , define the probability y_i belongs to class 1 as

$$\hat{p}_i = \frac{1}{1 + e^{-w^T x_i + b}}$$

Then, minimizes cross-entropy loss:

$$\ell(y, \hat{p}) = \sum_i [y_i \log(\hat{p}_i) + (1 - y_i)(1 - \hat{p}_i)]$$

- By converting X_i s to $\hat{p}_i$ s, reduces influence of x_i s that are outliers/high leverage points

10 Classification

- **Binary Classification:** Train model on weights w , bias b s.t. the decision boundary $w^T x + b$ maximizes correct predictions.
- **Confusion Matrix:**

Predicted	Actual	
	T	N
T	TP	FP
N	FN	TN

10.1 Multiclass Classification

- Instead of $y_i \in \{0, 1\}$, now $y_i \in \mathcal{L}$, where $\mathcal{L} = \{1, \dots, k\}$.
- Models for multiclass classification: Logistic Regression, NN/CNN with Cross Entropy loss

Metrics:

- **Accuracy:** $\frac{TP + TN}{TP + FP + TN + FN}$
- **Precision:** $\frac{TP}{TP + FP} = \frac{\text{True positives}}{\# \text{ of predicted positives}}$
 - How accurate are the model's positive classifications
- **Recall:** $\frac{TP}{TP + FN} = \frac{\text{True positives}}{\# \text{ of actual positives}}$
 - What is detection rate for the true positives!

Example (Multiclass):

Actual	Predicted		
	1	2	3
1	n_{11}	n_{12}	n_{13}
2	n_{21}	n_{22}	n_{23}
3	n_{31}	n_{32}	n_{33}

$$\text{Precision of class 2 : } \frac{n_{22}}{n_{12} + n_{22} + n_{32}}$$

$$\text{Recall of class 2 : } \frac{n_{22}}{n_{21} + n_{22} + n_{23}}$$

11 Two Sample Z-Test

$H_0 : \mu_A = \mu_B$ There is no difference in the two populations.

$H_A : \mu_A \neq \mu_B$ There is a difference in the two populations.

$$z^* = \frac{\bar{x}_A - \bar{x}_B - 0}{\sqrt{\frac{\sigma_A^2}{n_A} + \frac{\sigma_B^2}{n_B}}}$$

If $P(z > z^*) < \alpha$, reject H_0 for H_A .

- P-value is the false discovery rate.

12 Efficiency vs. Robustness

- **Efficiency** - tries to use as much of the training data as possible to reach a conclusion
- **Robustness** - protects against bad assumptions, noise, or inaccurate data
- Regularization is an example of robustness. Ridge regression, for example imposes prior $w \sim N(0, \frac{1}{\lambda})$.
 - Trade off a small increase in bias for a large decrease in variance
- Other robustness methods:
 - Acquire more data to decrease leverage of an influential point
 - Convert data to percentiles or ranks (Low Rank Prior)
 - * Can lose some of the data's meaningfulness, but improves robustness
 - Find a low-rank representation of data (PCA)

13 PCA

- Rank k dimension reduction given by
$$X_k = U_k \Sigma_k V_k^T$$
- Reduce high-dimensional data along a low-dimensional basis composed of principal components (PCs).
 - Each PC is in the direction of greatest variance in the data
- To find which basis is the result of PCA check:
 - Does the basis vector have stronger components in variance maximizing areas.
 - Does the basis keep positive/negative correlations as exhibited in the data?

14 Neural Networks

14.1 Forward Propagation

- Given input layer $X \in \mathbb{R}^{n \times p}$, forward propagation is defined by:
- At X (layer 0):

$$Z^{(0)} = W^{(0)}X + b^{(0)}, \quad A^{(0)} = \sigma(Z^{(0)})$$

- At hidden layer ℓ :

$$Z^{(\ell)} = W^{(\ell)}A^{(\ell-1)} + b^{(\ell)}, \quad A^{(\ell)} = \sigma(Z^{(\ell)})$$

- Then, $\hat{Y} = A^{(L)}$, and the loss function is given by $\ell(Y, \hat{Y})$ at layer L .

14.2 Back Propagation

$$\begin{aligned}
 \frac{\partial \ell}{\partial W^{(\ell)}} &= \frac{\partial \ell}{\partial A^{(\ell)}} \cdot \frac{\partial A^{(\ell)}}{\partial Z^{(\ell)}} \cdot \frac{\partial Z^{(\ell)}}{\partial W^{(\ell)}} \\
 &= \left[\frac{\partial \ell}{\partial A^{(\ell)}} \odot \sigma'(Z^{(\ell)}) \right] (A^{(\ell-1)})^T \\
 \\
 \frac{\partial \ell}{\partial A^{(\ell)}} &= \frac{\partial}{\partial A^{(\ell)}} \ell(\hat{y}, y) = \frac{\partial}{\partial A^{(\ell)}} \ell(A^{(L)}, y) = \frac{\partial}{\partial A^{(\ell)}} \ell \left(\sigma(\underbrace{W^{(L-1)} A^{(L-1)} + b^{(L-1)}}_{Z^{(L)}}), y \right) \\
 &= \frac{\partial}{\partial A^{(\ell)}} \ell \left(\sigma(W^{(L-1)} \underbrace{\sigma(W^{(L-2)} A^{(L-2)} + b^{(L-2)})}_{Z^{(L-1)}} + b^{(L-1)}), y \right) \\
 &\quad \dots
 \end{aligned}$$

Notes:

- $X_1, \dots, X_N$ should be standardized to prevent exploding/vanishing gradients
- One forward + backward pass is called an **epoch**
- With no activation function, we have that

$$\hat{y} = W^{(L-1)} W^{(L-2)} \dots W^{(1)} X + b^{(1)} + \dots + b^{(L-2)} + b^{(L-1)}$$

which is just a linear model.

```
from sklearn.preprocessing import StandardScaler
X_norm = StandardScaler().fit_transform(X)
```

14.3 Label Encoding, One-Hot Encoding

Label Encoding

```
from sklearn.preprocessing import LabelEncoder
X[label_encoded_var] = LabelEncoder().fit_transform(X[cat_col])
```

- Assigns each categorical variable outcome a numerical level.
- As such, label encoding assumes that there is an ordering in outcomes, when none may exist. This may make interpretation difficult, and may result in inflated coefficients.

One-Hot Encoding

```
from sklearn.preprocessing import OneHotEncoder
encoded_X_cols = OneHotEncoder().fit_transform(X[cat_col])
```

- Assigns each categorical variable outcome a separate column (or indicator variable).
- Let l be an arbitrary outcome. If $\mathbf{1}_{\{x_i \in l\}}$, then for the column associated with l , x_i 's value is 1.
- Unlike label encoding, one-hot encoding does not assume an ordering in outcomes.
- However, too many outcomes can lead to a significant increase in predictors, which can distort possible association and pull coefficients towards 0.

15 Convolution

- X is the input signal
- Y is the filter/kernel that slides along X

Ex. 1: $X = [a, b, c, d, e]$, $Y = [-1, 1]$

$$X * Y = [-a + b, -b + c, -c + d, -d + e]$$

All convolutions in this case are stride 1, no padding.

Ex. 2:

$$X = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad Y = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$X * Y = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

A convolution is a fast way to disseminate information throughout the network.

Ex. 3: Given $X = [a \ b \ c \ d \ e]$, $f_1 = [f_{11} \ f_{12}]$, $f_2 = [f_{21} \ f_{22}]$

$$f_1 * X = [f_{11}a + f_{12}b, \quad f_{11}b + f_{12}c, \quad f_{11}c + f_{12}d, \quad f_{11}d + f_{12}e]$$

$$\begin{aligned} f_2 * f_1 * X &= [f_{21}(f_{11}a + f_{12}b) + f_{22}(f_{11}b + f_{12}c), \\ &\quad f_{21}(f_{11}b + f_{12}c) + f_{22}(f_{11}c + f_{12}d), \\ &\quad f_{21}(f_{11}c + f_{12}d) + f_{22}(f_{11}d + f_{12}e)] \end{aligned}$$

- In each component in $f_2 * f_1 * X$, there are three data points (a, b, c) , (b, c, d) , (c, d, e) .
- But, the data points at the edges (a, e) are represented less than data points close to the middle.
- Each subsequent convolution reduces each dimension by 1

$$- \text{ Given } X \in \mathbb{R}^{n \times m}, X * Y \in \mathbb{R}^{(n-1) \times (m-1)}$$

16 Attention

- Given $X \in \mathbb{R}^d$, let $g \in \mathbb{R}^d$ be an attention filter of the same dimension.
- Then, each component of g_i determines how much attention is placed on the component X_{ij} .

$$Xg = \begin{bmatrix} -x_1- \\ \vdots \\ -x_n- \end{bmatrix} \begin{bmatrix} g_1 \\ \vdots \\ g_d \end{bmatrix} = \begin{bmatrix} x_{11} \cdot g_1 & x_{12} \cdot g_2 & \dots & x_{1d} \cdot g_d \\ \vdots & & & \vdots \\ x_{n1} \cdot g_1 & x_{n2} \cdot g_2 & & x_{nd} \cdot g_d \end{bmatrix}$$

17 Large Language Models

- LLMs are fundamentally next-token predictors
- Each word is assigned a unique token t
- Then, given tokens $t_1 \dots t_k$, predict t_{k+1}

LLMs are trained in the following way:

1. **Foundational Model Training** - learn weights from entire Internet
2. **Mid-training** - Refine weights on questions asked on the Internet, learn a small set of conversational data
3. **Post-training** - Using real people to finetune the model with Sophisticated Conversations

LLMs typically downsample user queries into a densely represented embedding vector. This embedding vector can then be compared to saved embeddings in a vector database, and can further

- **Encoder/Autoencoder** - reduce $X_1 \dots X_n$ to dense representation (embedding vector), then upsample the response.
- Embeddings can be stored in vector database. Pre-computed embeddings by a well-trained model can enhance the response of other LLMs that query the vector database. The vector database can also serve as an embedding cache.